



Amrita School of Computing, Amritapuri Campus
Department of Computer Science and Engineering

23CSE498

Project Phase II

2026

B.Tech CSE — Initial Project Idea Submission

Project Team

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Guide Details

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1. Project Title

Project Title:

Semi-Supervised Learning for Cross-Dataset Generalization in Cervical Cytology Image Classification

2. Project Abstract

2.1 Motivation

Cervical cancer remains one of the leading causes of mortality in women due to cancers, especially in low- and middle-income nations due to the unavailability of expertise for cytopathology. Pap smear testing is an efficient way to detect precancerous changes, but the procedure is highly tedious, time-consuming, and is subject to observer variation. The application of deep learning has proven promising in this area, though such models need lots of expert-labeled images.

In order to address the shortage of annotated data, semi-supervised learning (Semi-SL) has been suggested as an effective approach by leveraging the huge amount of unlabeled data. Yet, current Semi-SL algorithms in the context of cervical cytology are tested on single data sets with the assumption that the training and test images are coming from the same data distribution. In the real application, images coming from different hospitals, scanned by different machines and stained by different processes have distribution shift and thus can cause serious drops in performance (for example, Coskun et al. (2026) show the accuracy drops from $\sim 95\%$ to $\sim 72\%$ when transferring across different scanners using supervised learning). It becomes critical to know whether the unlabeled data can be used to mitigate the distribution shift issue, especially because collecting large-scale labeled data sets for each hospital becomes unrealistic.

2.2 Problem Statement

The current semi-supervised approaches for cervical cytology classification have concentrated only on optimizing the performance of the method in a single dataset with a small amount of labeled samples. These approaches do not look at how well the representations obtained through few labeled and many unlabeled samples can generalize well across independently obtained datasets with varying stain protocol, imaging instruments, and patients. Thus, it is unclear how well the current Semi-SL approaches perform when subjected to changes in domain in the real world. The current work is driven by the following research question: Will semi-supervised learning help, hinder, or have no effect on cross-dataset generalization in cervical cytology? The objective of this work is to explore the benefits of Semi-SL approaches for improving cross-dataset generalization in cervical cytology image classification.

Scope note: Considering the B.Tech schedule, the core experiments will be performed on pairwise transfer of the main datasets (CRIC $\leftrightarrow$ RIVA) through representative semi-supervised learning setups in 5% and 10% labeling schemes. The other datasets (LBC) and the 1% labeling scheme are the stretch goals for future work.

3. Objectives

1. **Align** the two major datasets (CRIC Cervix, RIVA) into a unified binary classification problem set-up (Normal/Abnormal) with standardized pre-processing steps.
2. **Develop** and **test** state-of-the-art approaches for semi-supervised learning in which the unlabeled data comes from the same training dataset as the labeled data in terms of a small label fraction scenario (5% and 10%).
3. Test the **generalization ability** of each of these models via a **pairwise transfer setting** (Train on Dataset A $\rightarrow$ Test on Dataset B). This follows the protocol of Coskun et al. (2026).
4. **Compare** the performance of the semi-supervised approaches against fully-supervised base-

lines under equivalent cross-dataset evaluation setups. The performance will be measured via accuracy, macro F1-score, precision, recall, and AUROC.

5. **Examine** the impact of the domain shift specific to each dataset on the classification performance in supervised and semi-supervised scenarios.

4. Dataset / Data Source

The project will use publicly available cervical cytology datasets to conduct a cross-dataset generalization study.

1. CRIC Cervix Dataset (Primary)

- **Source:** Figshare / Scientific Data
- **Link:** https://figshare.com/articles/dataset/CRIC_Cervix/13263014
- **Details:** Conventional Pap smear, containing 11,534 annotated cervical cells from 118 patients.

2. RIVA Dataset (Primary)

- **Source:** Zenodo / Scientific Data (Nature Portfolio)
- **Link:** <https://zenodo.org/records/13979911>
- **Details:** Conventional Pap smear cytology, 115 patients, across 8 diagnostic categories.

3. LBC Dataset (Secondary, if time permits)

- **Source:** Mendeley Data
- **Link:** <https://data.mendeley.com/>
- **Details:** Liquid-Based Cytology (LBC), 963 cervical images from 460 patients.

Note: Dataset selection is tentative and may be updated based on availability and guide suggestions.

5. Expected Deliverables

1. **Code Repository:** A well-implemented Python code for the cross-dataset evaluation framework, the data loader code, and the Semi-SL training code.
2. **Benchmark Results:** The performance charts and tables comparing the Semi-SL approach to the supervised baseline method for all data transfer settings and all label ratios.
3. **Research Paper:** A complete paper for submission to an internationally reputed conference (such as IEEE INDICON, ICPR).

6. Preliminary Literature Survey

No.	Paper Title	Year	Key Contribution
1	SCAC: A Semi-Supervised Learning Approach for Cervical Abnormal Cell Detection (IEEE JBHI)	2024	Introduces a teacher-student Semi-SL framework with Swin Transformer and augmentation strategies specifically for cervical abnormal cell detection.
2	A Semi-supervised Deep Learning Method for Cervical Cell Classification (Analytical Cellular Pathology)	2022	Proposes an iterative pseudo-labeling and classifier fusion framework for cervical cell classification using limited labeled data.
3	Accurate Deep Learning Model using Semi-supervised Learning and Noisy Student for Cervical Cancer Screening (PLOS ONE)	2023	Applies the Noisy Student semi-supervised paradigm to low-magnification cervical cancer screening WSI tiles, improving screening accuracy.
4	Self-Path: Self-Supervision for Classification of Pathology Images with Limited Annotations (IEEE TMI)	2021	Introduces pathology-specific self-supervised pretext tasks and demonstrates improved domain adaptation across histopathology datasets.
5	Robust Cell-Level Classification for Liquid-Based Cervical Cytology Using Deep Transfer Learning: A Multi-Source Study Addressing Scanner-Induced Domain Shifts (Bioengineering)	2026	Empirically demonstrates severe performance degradation ($\sim 95\% \rightarrow \sim 72\%$) in cervical cytology when transferring models across scanners/datasets, but uses only supervised learning.

6.1 Novelty / Research Gap Statement

Novelty / Research Gap:

Supervised experiments recently conducted by Coskun et al. (2026), have revealed that cervical cytology classifiers suffer large reductions in performance, such as over 20% reduction in accuracy, when evaluated against data coming from other scanners or other organizations, under the condition of pairwise transfer. On the other hand, the works of SCAC, Zhao, and Kurita related to semi-supervised learning (Semi-SL) have extensively proven the efficiency of Semi-SL in reducing annotation efforts. Nevertheless, one problem arises – Semi-SL techniques currently applied are limited to being assessed using held-out partitions from the same source distribution, ignoring the problems mentioned by Coskun et al.

The effects of Semi-SL on the ability to generalize across various datasets for cervical cytology remain unknown. This project will fill the knowledge gap by examining the use of appropriate semi-supervised learning approaches through cross-dataset transfer. Following the study conducted by Coskun et al. in supervised domain-shift learning, our study will test the transfer pair approach (Train on one dataset $\rightarrow$ Test on another) to find out if Semi-SL can handle or is impacted by domain shift. The proposed study, which uses public cervical cytology datasets, will give a clear testing framework for cross-dataset cervical cancer screening..

6.2 Comparison of Related Work

No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
1	Zhang et al. (SCAC) [1]	2024	Teacher-Student Semi-SL with Transformer	Tianchi, ComparisonDetector	No cross-dataset evaluation; all experiments within single source datasets.
2	Zhao et al. [2]	2022	Pseudo-label Semi-SL + Classifier Fusion	Private In-house, Herlev	Single dataset evaluation; does not assess generalization to independent cytology datasets.
3	Kurita et al. [3]	2023	Noisy Student Semi-SL	Private Single-Institution	No cross-domain evaluation; model trained and tested on slides from a single hospital scanner.
4	Koohbanani et al. (Self-Path) [4]	2021	SSL + Domain Adaptation	Histopathology Datasets	Not cervical cytology; evaluated on breast/colorectal tissue, not Pap smear cells.
5	Coskun et al. [5]	2026	Supervised Transfer Learning (Pairwise)	Herlev, SIPaKMeD, CRIC	Supervised only; proves pairwise cross-dataset shift exists but does not utilize semi-supervised learning to mitigate it.

Table 1: Comparison of related work across key aspects

7. Potential Publication Venues

Possible Conferences

- **IEEE INDICON** (India) — Reputable IEEE conference held in India, suitable for presenting AI applications in healthcare.
- **ICPR** (International Conference on Pattern Recognition) — Strong fit for the computer vision and pattern classification components of the work.
- **IEEE ICACCS** (India) — IEEE conference on advanced computing and communication systems; relevant for applied machine learning in healthcare.

Possible Journals

- **Computer Methods and Programs in Biomedicine (Elsevier)** — well-suited for applied, reproducible AI pipelines in biomedical imaging.

- **Scientific Reports (Nature)** — broad-scope journal that publishes well-executed empirical studies.
- **IEEE Access** — open-access journal with a strong engineering focus; suitable for applied AI in healthcare.

8. References

- [1] Z. Zhang *et al.*, “SCAC: A Semi-Supervised Learning Approach for Cervical Abnormal Cell Detection,” *IEEE J. Biomed. Health Inform.*, vol. 28, no. 6, pp. 3501–3512, Jun. 2024, doi: 10.1109/JBHI.2024.3375889.
- [2] S. Zhao, Y. He, J. Qin, and Z. Wang, “A Semi-supervised Deep Learning Method for Cervical Cell Classification,” *Anal. Cell. Pathol.*, vol. 2022, Art. no. 4376178, 2022, doi: 10.1155/2022/4376178.
- [3] Y. Kurita *et al.*, “Accurate deep learning model using semi-supervised learning and Noisy Student for cervical cancer screening in low magnification images,” *PLoS ONE*, vol. 18, no. 5, Art. no. e0285996, May 2023, doi: 10.1371/journal.pone.0285996.
- [4] N. A. Koohbanani, B. Unnikrishnan, S. A. Khurram, P. Krishnaswamy, and N. Rajpoot, “Self-Path: Self-Supervision for Classification of Pathology Images With Limited Annotations,” *IEEE Trans. Med. Imag.*, vol. 40, no. 10, pp. 2845–2856, Oct. 2021, doi: 10.1109/TMI.2021.3056023.
- [5] G. Coskun, M. C. Akuner, and E. Kaplanoglu, “Robust Cell-Level Classification for Liquid-Based Cervical Cytology Using Deep Transfer Learning: A Multi-Source Study Addressing Scanner-Induced Domain Shifts,” *Bioengineering*, vol. 13, no. 3, Art. no. 289, Feb. 2026, doi: 10.3390/bioengineering13030289.

9. Faculty Remarks

As guide, I hereby confirm that I checked the plagiarism and AI content.

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